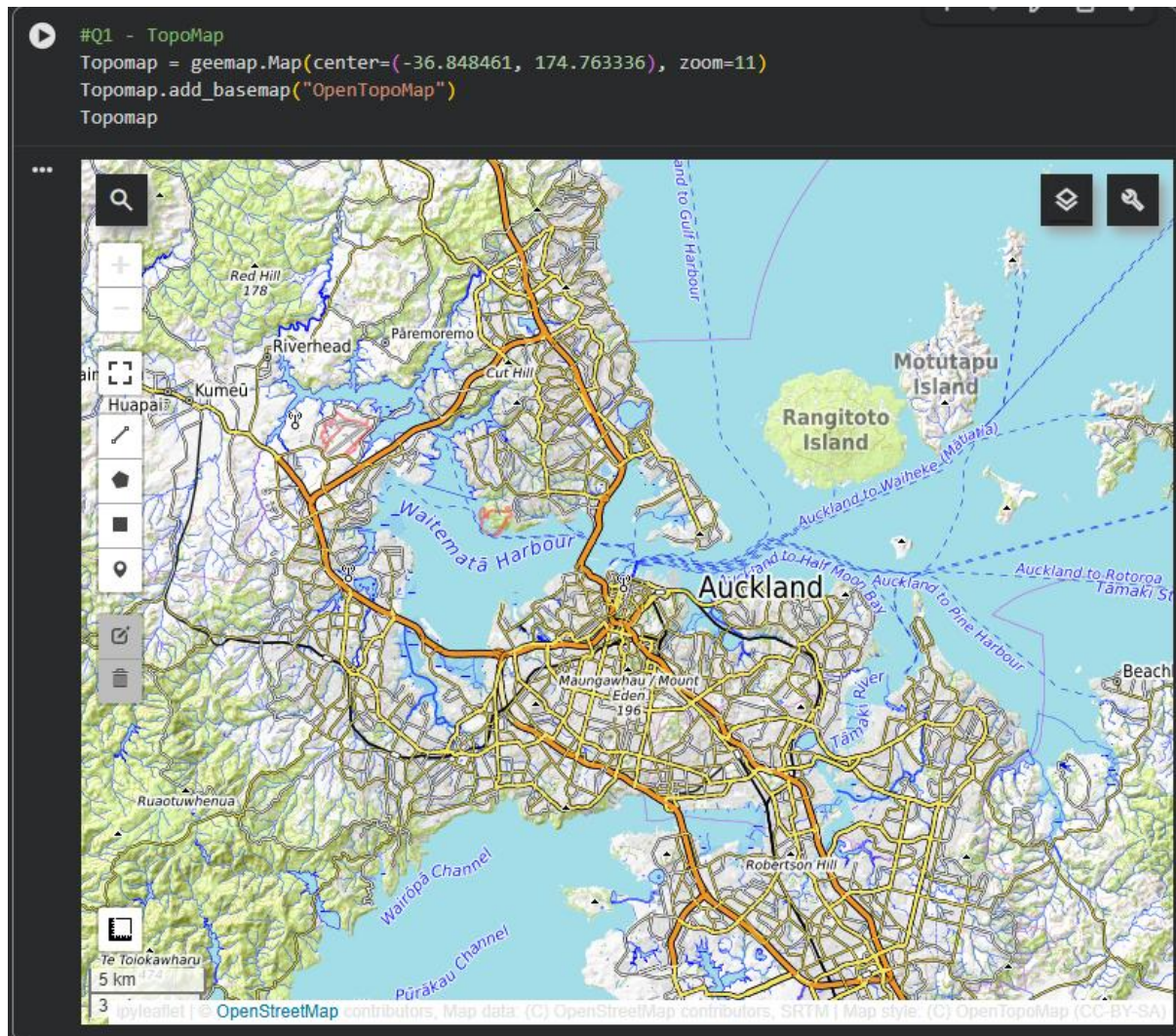


Answer Proforma

(1) Find and add the basemap 'OpenTopoMap' to the display window AND change the code so that the map opens and zooms to the city of Auckland. Provide a screenshot of your code and the output. (2 pts)



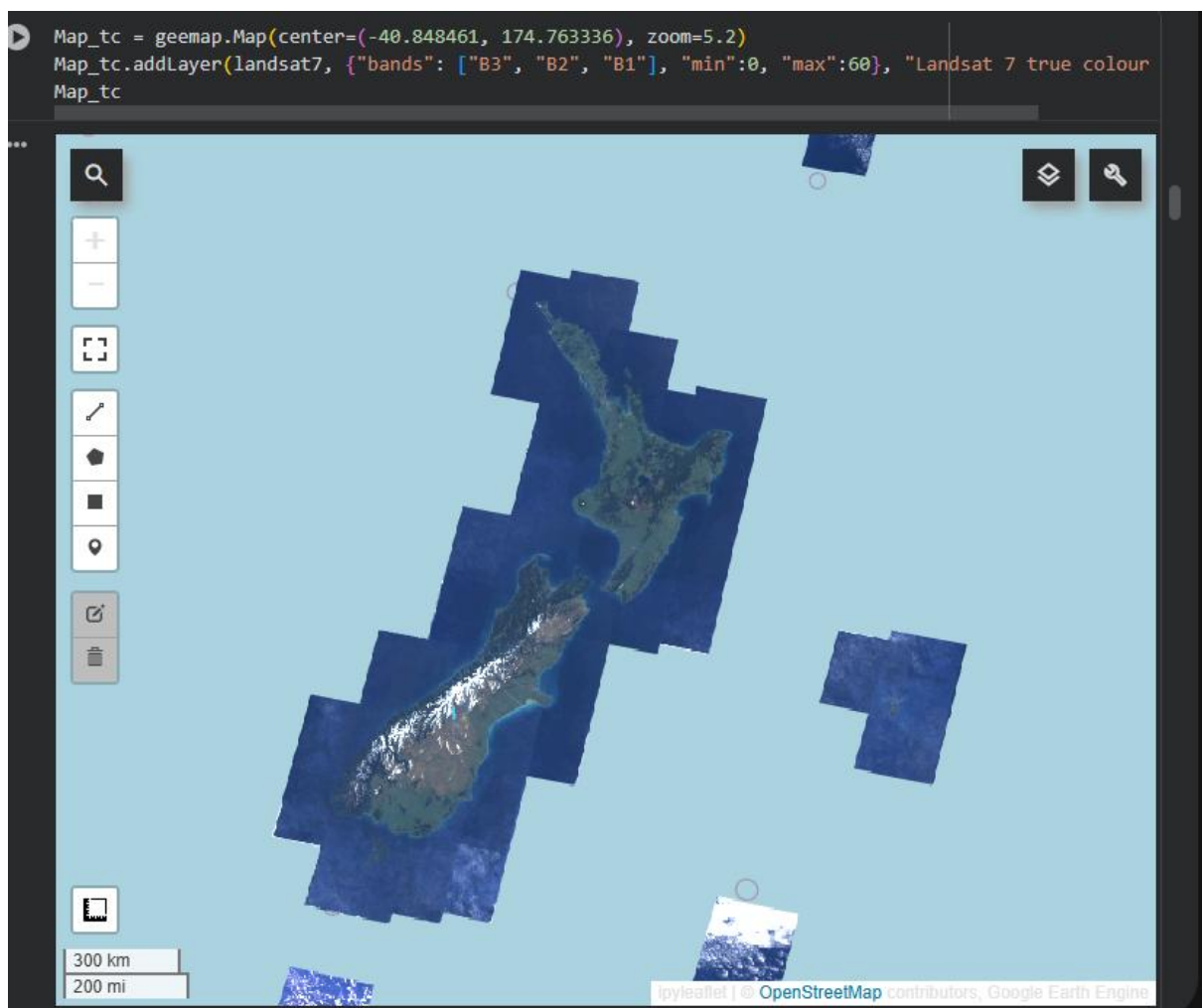
(2) The Landsat 7 data displays as red colored/tinged image. Why is this? [Think about both the content in Lecture 1, and what is going on when you visualize a satellite image]. (2 pts)

We are calling for the bands: B4 (Near-IR), B3 (red), and B2 (green). These bands capture spectral reflectance in their respective EM wavelengths, producing an image composite composed of all three bands. Land in the image is largely displayed as red due to the presence of vegetation which reflects Near-IR light. The intensity of the colours shown is dependent on

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the spectral reflectance of each individual pixel. The outputted intensity of pixels can be adjusted by altering the min and max values in the band visualisation to change contrast if desired.

(3) Correct the display of the Landsat 7 image so that it displays in 'True Colour'. Provide a screenshot of your code and the corrected output over the whole of New Zealand. (10 pts)



(4) Why are we using the surface reflectance (SR) here rather than the top of atmosphere (ToA)? (2 pt)

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SR data has been atmospherically corrected so that the reflectance data is representative of the ground values without the interference from the atmosphere. This is ideal for capturing spectral reflectance values as we are looking for the actual spectral characteristics of our target on the ground. Atmospheric interference may cause misclassification/misidentification in ML, and also reduces the validity of comparison.

(5) What is a 'scale factor' as has been used in the prior code cell, and why are they used in satellite data storage-to-processing workflows such as this? (2 pts)

The scale factor is acting to decompress the stored sensor data from its digital number format (integer) to its actual values via a multiplication. Data is compressed into 16-bit integers to reduce required storage and data transfer time.

(6) If we want to apply multiple reducers to the same inputs what is a more efficient way to do so than just writing out a long series of sequential reducers on our end? (2 pts)

`ee.Reducer.combine()` can be used to perform multiple reductions, such as the combination of mean and stddev reducers (GEE python example code):

```
# Example code from GEE Guidelines
# Load a Landsat 8 image.
image = ee.Image('LANDSAT/LC08/C02/T1/LC08_044034_20140318')

# Combine the mean and standard deviation reducers.
reducers = ee.Reducer.mean().combine(
    reducer2=ee.Reducer.stdDev(), sharedInputs=True
)

# Use the combined reducer to get the mean and SD of the image.
stats = image.reduceRegion(reducer=reducers, bestEffort=True)

# Display the dictionary of band means and SDs.
display(stats)
```

► Object (34 properties)

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(7) Produce a table that summarises the amount of cloud cover per year (as a percentage) for 2005, 2010 and 2015 over a region of your choice. Provide a screenshot of the code that does so as well as your output table. (5 pts)

This is a long code section because I produced a formatted table with matplotlib:

```
import pandas as pd

# Lake Geneva region
lake_geneva = ee.Geometry.Rectangle([5.8, 46.2, 6.7, 46.6])

# function to map over years
def process_year(year):
    year = ee.Number(year)
    start_date = ee.Date.fromYMD(year, 1, 1)
    end_date = ee.Date.fromYMD(year, 12, 31)

    col = (ee.ImageCollection('LANDSAT/LE07/C02/T1_L2')
           .filterBounds(lake_geneva)
           .filterDate(start_date, end_date))

    # call to return features
    return ee.Feature(None, {
        'Year': year,
        'Satellite': 'Landsat 7 ETM+',
        'Scene Count': col.size(),
        'Average Cloud Cover (%)': col.aggregate_mean('CLOUD_COVER')
    })

# compute across each year
years_list = ee.List([2005, 2010, 2015])
results_fc = ee.FeatureCollection(years_list.map(process_year))

# fetch results
raw_data = results_fc.getInfo()
# parse into a DF
table_data = [feat['properties'] for feat in raw_data['features']]
df_results = pd.DataFrame(table_data)

# set column names
df_results = df_results[['Year', 'Satellite', 'Scene Count', 'Average Cloud Cover (%)']]
# round to 2dp
df_results['Average Cloud Cover (%)'] = df_results['Average Cloud Cover (%)'].round(2)

# Generate Figure Table
fig, ax = plt.subplots(figsize=(10, 3), dpi=300)
ax.axis('off')
ax.axis('tight')

# using matplotlib for the table
table = ax.table(
    cellText=df_results.values,
    colLabels=df_results.columns,
    cellLoc='center',
    loc='center',
    colColours=['#e0e0e0'] * len(df_results.columns) #background colour
)

# format layout and font
table.auto_set_font_size(False)
table.set_fontsize(12)
table.scale(1.2, 2.0)

plt.title("Landsat 7 Cloud Cover Statistics over Lake Geneva (2005, 2010, 2015)",
          fontsize=14, fontweight='bold', pad=20)
plt.tight_layout()
plt.show()
```

Landsat 7 Cloud Cover Statistics over Lake Geneva (2005, 2010, 2015)

Year	Satellite	Scene Count	Average Cloud Cover (%)
2005	Landsat 7 ETM+	33	35.76
2010	Landsat 7 ETM+	44	34.09
2015	Landsat 7 ETM+	93	42.4

(8) Using Sentinel 2, create a figure that does the following over the city of Christchurch, NZ:

- **Conduct a basic quality assessment of the data you use, using a cloudy pixel % estimate.**
- **Describe/show the spectral reflectance signatures of: urban, forest, grassland, water and bare soil.**
- **Illustrates the sample areas that you have used to create these spectral signatures.**

Provide a publication quality copy of this figure in the answer proforma. Publication quality means that:

- 1. It is at a good quality (resolution) and not blurry.**
- 2. The font size of all elements is big enough to read easily on a screen when placed in the document.**
- 3. Has a figure caption that ensures the reader knows what you have done and why.**
- 4. Tells the story 'at a glance'. Don't make your viewer work too hard to understand your core message.**

(25 pts)

(Next Page)

